

CS 171: Discussion 2 (Sept 8)

1. Equivalence of Definitions

You are given an encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ that satisfies the condition

$$\Pr[M = m | C = c] = \Pr[M = m]$$

for every probability distribution M over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ such that $\Pr[C = c] > 0$. Show that for any two messages $m, m' \in \mathcal{M}$ and for any ciphertext $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c].$$

2. Non-Negligible Function

A function $f : \mathbb{Z}^+ \rightarrow [0, 1]$ is a *negligible function* if $\forall$ polynomials $p(\cdot)$, $\exists N \in \mathbb{Z}^+$ such that $\forall n > N$ we have $f(n) < \frac{1}{p(n)}$.

See the section on *negation* at https://en.wikipedia.org/wiki/Universal_quantification. Define a non-negligible function using the negation of the definition of a negligible function.

3. A Different One-time Pad

For set S , let $s \xleftarrow{\$} S$ denote selecting a uniformly random element s from S . Consider the following encryption scheme for the message space $\{0, 1\}$.

- **Gen**: Choose two random bits $a, b \xleftarrow{\$} \{0, 1\}$. Output (a, b) .
 - **Enc** $((a, b), m)$: Choose random $x_1 \xleftarrow{\$} \{0, 1\}$ and compute x_2 such that $a \cdot x_1 + b + x_2 = m$ where $+$ and $\cdot$ are operations over $\mathbb{Z}_2$, aka the integers mod 2. Output (x_1, x_2) .
 - **Dec** $((a, b), (x_1, x_2))$: Output $m = a \cdot x_1 + b + x_2$.
- (a) Show that this scheme is perfectly secure. *Hint: Use the second (equivalent) definition of perfect secrecy from Q1.*
- (b) Recall that the one-time pad you saw in class can be broken under a *known-plaintext attack*: upon receipt of both a plaintext and corresponding ciphertext for a message, the attacker can recover information about the key.
- (i) Is the scheme above also susceptible to a known-plaintext attack?
 - (ii) For the scheme from class, how many plaintext-ciphertext pairs are necessary to recover the entire key? How about for this new scheme—or is it impossible to fully recover the key, even with unlimited plaintext-ciphertext pairs?